GED Math Practice Test 12

PART A — QUESTIONS

GED-MATH12-001
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A school cafeteria packed 5 trays with 16 sandwiches on each tray. If 14 sandwiches were set aside for staff, how many sandwiches were left for students?
A) 54
B) 66
C) 74
D) 94

GED-MATH12-002
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
What is the value of $4x^2 - 3x$ when $x = 3$?
A) 21
B) 27
C) 33
D) 45

GED-MATH12-003
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A tank is 5/6 full. Then 1/3 of the tank is drained. What fraction of the tank remains full?
A) 1/2
B) 2/3
C) 3/4
D) 7/9

GED-MATH12-004
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
Which expression is equivalent to $6 - 4(2n - 3)$?
A) $-8n - 6$
B) $-8n + 18$

C) $8n - 6$
D) $8n + 18$

GED-MATH12-005
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A box contains 45 folders. If 18 folders are yellow, what percent of the folders are yellow?
A) 30%
B) 36%
C) 40%
D) 45%

GED-MATH12-006
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $A = 2(x + 5)^2$ gives the area A of a square-based display section. If $A = 98$, what is one possible value of x?
A) 2
B) 4
C) 7
D) 12

GED-MATH12-007
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A hospital supply order includes 22 cartons. Each carton contains 15 packs, and each pack has 8 sterile pads. How many sterile pads are in the order?
Type your answer as an integer.

GED-MATH12-008
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $P = 2a + 3b$ can be used to calculate points from two task types. To solve for b, first subtract [D1] from both sides, then divide by [D2].
Dropdowns:
D1) 2a | 3b | P
D2) 2 | 3 | a

GED-MATH12-009
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A water tank held 96 gallons at the start of the day. A hose added 3.5 gallons per minute for 12 minutes, and then 18 gallons were used. How many gallons were in the tank after these changes?
A) 108
B) 120
C) 132
D) 156

GED-MATH12-010
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A value of t satisfies $4(t - 3) \leq 28$. Which statement describes all possible values of t?
A) $t \leq 4$
B) $t \leq 10$
C) $t \geq 10$
D) $t \geq 31$

GED-MATH12-011
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A table shows production data for three work teams.

Team: A, B, C
Hours worked: 5, 6, 8
Units completed: 155, 198, 240

Which team completed the most units per hour?
A) Team A
B) Team B
C) Team C
D) Teams A and C were tied

GED-MATH12-012
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
Which expressions are equivalent to $7(3x - 2) - 4(x + 5)$?

A) $17x - 34$
B) $21x - 14 - 4x - 20$
C) $17x + 6$
D) $7(3x) - 7(2) - 4x - 20$
E) $3(7x - 6)$

GED-MATH12-013
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A training program recorded how participants completed a lesson.

Completion method: Laptop, Phone, Tablet, Printed packet
Participants: 38, 27, 15, 20

If one participant is selected at random, what is the probability that the participant used either a phone or a tablet?
A) 21/50
B) 27/100
C) 3/10
D) 7/20

GED-MATH12-014
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A formula for the total number of points earned in a course is $P = 12q + 8a$, where q is the number of quizzes and a is the number of assignments. If $P = 252$ and $a = 18$, what is q?
Type your answer as an integer.

GED-MATH12-015
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A storage area is made from a rectangle with a triangular extension on one side. The rectangle is 18 feet by 11 feet. The triangle has a base of 11 feet and a height of 8 feet. What is the total area?
A) 154 square feet
B) 198 square feet
C) 242 square feet
D) 286 square feet

GED-MATH12-016
Section: Math
Type: mcq
Category: Algebraic Problem Solving

Calculator: Calculator-Allowed
Prompt:
The table shows part of a linear relationship.

x: −4, 2, 5, 11
y: 31, 13, 4, −14

Which equation represents the relationship?
A) $y = -2x + 23$
B) $y = -3x + 19$
C) $y = 3x + 43$
D) $y = -6x + 7$

GED-MATH12-017
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
A store compares two bags of dog food.

Bag: Medium, Large
Weight: 18 lb, 32 lb
Price: $28.80, $48.00

Which statements are true?
A) The medium bag costs $1.60 per pound.
B) The large bag costs $1.50 per pound.
C) The large bag has the higher unit price.
D) The medium bag costs $1.50 per pound.
E) The two bags have the same unit price.

GED-MATH12-018
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which inequality represents "three times the sum of a number and 4 is at least 45"?
A) $3x + 4 \geq 45$
B) $3(x + 4) \geq 45$
C) $3(x - 4) \geq 45$
D) $x + 3(4) \geq 45$

GED-MATH12-019
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:

A store sold an item for \$184 after a 15% markup from its wholesale cost. What was the wholesale cost?
A) \$156.40
B) \$160.00
C) \$169.00
D) \$211.60

GED-MATH12-020
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for x: $3(5x − 4) − 2(4x + 7) = 58$.
Type your answer as an integer.

GED-MATH12-021
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A circular tabletop has an area of about 78.5 square feet. Using $\pi \approx 3.14$, what is the radius of the tabletop?
A) 4 feet
B) 5 feet
C) 10 feet
D) 25 feet

GED-MATH12-022
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A line passes through the points (−8, −6) and (4, 12). What is the rate of change of the line?
A) 1/2
B) 3/2
C) 2
D) 3

GED-MATH12-023
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A driver travels 64.8 miles in the first part of a route and 38.5 miles in the second part. The full route is 125 miles. How many miles remain?
A) 21.7
B) 22.3

C) 25.5
D) 28.7

GED-MATH12-024
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Function A is represented by the table.

x: −2, 1, 7, 10
y: −1, 8, 26, 35

Function B is represented by $y = 2.5x + 4$. Which statement is true?
A) Function A has a greater rate of change than Function B.
B) Function B has a greater rate of change than Function A.
C) Both functions have the same rate of change.
D) Function A is not linear.

GED-MATH12-025
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A certification score is based on 30% written work, 45% skills demonstration, and 25% attendance. A student scores 82 on written work, 90 on skills demonstration, and 96 on attendance. What is the certification score?
Type your answer as a decimal rounded to the nearest tenth.

GED-MATH12-026
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A theater sold 138 tickets. Balcony tickets cost $18 each, and floor tickets cost $26 each. The total revenue was $3,084. Which system can be used to find b, the number of balcony tickets, and f, the number of floor tickets?
A) $b + f = 3{,}084$
$18b + 26f = 138$
B) $b + f = 138$
$18b + 26f = 3{,}084$
C) $18b + 26f = 138$
$b - f = 3{,}084$
D) $26b + 18f = 138$
$b + f = 3{,}084$

GED-MATH12-027
Section: Math

Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular image is 12 inches wide and 5 inches tall. It is enlarged by a scale factor of 1.4. What is the area of the enlarged image?
A) 84.0 square inches
B) 96.0 square inches
C) 117.6 square inches
D) 168.0 square inches

GED-MATH12-028
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The equation $8x + 5y = 30$ can be rewritten as y = [D1]x + [D2].
Dropdowns:
D1) $-8/5$ | $8/5$ | $5/8$
D2) 6 | -6 | 30

GED-MATH12-029
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A container has 7 green beads, 5 red beads, and 8 clear beads. One bead is selected at random and not replaced. Then a second bead is selected. What is the probability that both beads are clear?
A) 7/95
B) 14/95
C) 2/5
D) 16/25

GED-MATH12-030
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The formula $C = 2\pi r + 2s$ gives the perimeter C of a shape made from a semicircular edge and two straight sides of length s. Which equations correctly solve for s?
A) $s = (C - 2\pi r)/2$
B) $s = C/2 - \pi r$
C) $s = C - 2\pi r/2$
D) $s = C/(2\pi r) - 2$
E) $s = 2(C - \pi r)$

GED-MATH12-031
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A metal part has a volume of 42 cubic centimeters and a density of 7.8 grams per cubic centimeter. What is its mass?
A) 5.4 grams
B) 49.8 grams
C) 327.6 grams
D) 420.0 grams

GED-MATH12-032
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for r: $6(2r - 5) - 3(r + 4) = 66$.
A) 8
B) 10
C) 12
D) 14

GED-MATH12-033
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A cone-shaped cup has a radius of 3 inches and a height of 7 inches. Using $\pi \approx 3.14$, what is the volume of the cup?
Use $V = 1/3\pi r^2h$.
A) 21.98 in^3
B) 65.94 in^3
C) 131.88 in^3
D) 197.82 in^3

GED-MATH12-034
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The expression $x^2 - 13x + 36$ is equal to 0 for two values of x. Which value is one solution?
A) 3
B) 4
C) 13
D) 36

GED-MATH12-035
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A piece of equipment is modeled by V = 5,200(0.875)^t, where t is the number of years after purchase. Which statement best describes the change each year?
A) The value decreases by 12.5%.
B) The value decreases by 87.5%.
C) The value increases by 12.5%.
D) The value increases by 87.5%.

GED-MATH12-036
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A scale drawing uses 6 centimeters to represent 27 feet. A wall is 14 centimeters long on the drawing. What is the actual length of the wall?
A) 54 feet
B) 60 feet
C) 63 feet
D) 72 feet

GED-MATH12-037
Section: Math
Type: drag_drop
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Match each target with the correct tile.
Tiles:
A) $y = 3(x - 4) + 8$
B) $y = 3x - 4 + 8$
C) $y = 8(x - 4) + 3$
D) $y = 3(x + 8) - 4$
Targets:
T1) Subtract 4 from a number, multiply by 3, then add 8
T2) Multiply a number by 3, subtract 4, then add 8
T3) Add 8 to a number, multiply by 3, then subtract 4

GED-MATH12-038
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A machine prints 156 labels in 24 minutes. At this rate, how many labels will it print in 70 minutes?
A) 385

B) 420
C) 455
D) 468

GED-MATH12-039
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A conference room can hold no more than 180 people. There are already 36 staff members in the room. Each invited group has 12 people. Which inequality can be used to find g, the number of invited groups that can enter?
A) $36 + 12g \leq 180$
B) $36 + 12g \geq 180$
C) $12 + 36g \leq 180$
D) $180 + 12g \leq 36$

GED-MATH12-040
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A population of cells is modeled by N = 240(1.18)^t. Which statement best describes the meaning of 1.18?
A) The population increases by 18% each time period.
B) The population decreases by 18% each time period.
C) The starting population is 118 cells.
D) The population increases by 1.18 cells each time period.

GED-MATH12-041
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The table shows the distance and time for two delivery routes.

Route: East, West
Distance: 96 mi, 135 mi
Time: 2.4 hr, 3.0 hr

Which statements are true?
A) The East route averaged 40 miles per hour.
B) The West route averaged 45 miles per hour.
C) The West route was slower than the East route.
D) The East route averaged 42 miles per hour.
E) Both routes had the same average speed.

GED-MATH12-042
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
For the function $f(x) = -2x^2 + 5x + 12$, what is $f(-2)$?
A) −18
B) −6
C) 10
D) 14

GED-MATH12-043
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows a relationship.

x: −1, 0, 1, 2
y: 9, 4, 1, 0

Which statement best describes the relationship?
A) It is linear because the x-values increase by 1.
B) It is linear because the y-values decrease.
C) It is not linear because the changes in y are not constant.
D) It is not linear because one y-value is 0.

GED-MATH12-044
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which expression represents 9 more than one fourth of the difference of a number x and 12?
A) $1/4x - 12 + 9$
B) $1/4(x - 12) + 9$
C) $9(x - 12)/4$
D) $1/4(x + 12) - 9$

GED-MATH12-045
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $A = 1/2(a + b)h$ gives the area of a trapezoid. If $A = 168$, $a = 14$, and $h = 8$, what is b?
Type your answer as an integer.

GED-MATH12-046
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular parking area is 54 feet long and 36 feet wide. A rectangular storage shed that is 12 feet by 9 feet is placed in one corner. What area of the parking area remains uncovered?
A) 108 square feet
B) 1,044 square feet
C) 1,836 square feet
D) 1,944 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH12-001 — Correct: B — Correct Answer: 66 — Explanation: The cafeteria packed $5 \times 16 = 80$ sandwiches. After 14 were set aside, $80 - 14 = 66$ remained.

GED-MATH12-002 — Correct: B — Correct Answer: 27 — Explanation: Substitute $x = 3$: $4(3^2) - 3(3) = 36 - 9 = 27$.

GED-MATH12-003 — Correct: A — Correct Answer: 1/2 — Explanation: Since $5/6 - 1/3 = 5/6 - 2/6 = 3/6$, the tank remains 1/2 full.

GED-MATH12-004 — Correct: B — Correct Answer: $-8n + 18$ — Explanation: $6 - 4(2n - 3) = 6 - 8n + 12 = -8n + 18$.

GED-MATH12-005 — Correct: C — Correct Answer: 40% — Explanation: 18 out of 45 folders are yellow. Since $18/45 = 0.40$, 40% are yellow.

GED-MATH12-006 — Correct: A — Correct Answer: 2 — Explanation: $98 = 2(x + 5)^2$, so $49 = (x + 5)^2$. One solution is $x + 5 = 7$, giving $x = 2$.

GED-MATH12-007 — Correct: 2640 — Tolerance: exact — Correct Answer: 2640 — Explanation: The order contains $22 \times 15 \times 8 = 2{,}640$ sterile pads.

GED-MATH12-008 — Correct: D1=2a; D2=3 — Correct Answer: D1=2a; D2=3 — Explanation: From $P = 2a + 3b$, subtract $2a$ to get $P - 2a = 3b$, then divide by 3.

GED-MATH12-009 — Correct: B — Correct Answer: 120 — Explanation: The hose adds $3.5 \times 12 = 42$ gallons. Then $96 + 42 - 18 = 120$ gallons.

GED-MATH12-010 — Correct: B — Correct Answer: $t \leq 10$ — Explanation: Divide $4(t - 3) \leq 28$ by 4 to get $t - 3 \leq 7$. Add 3 to get $t \leq 10$.

GED-MATH12-011 — Correct: B — Correct Answer: Team B — Explanation: Team A: $155 \div 5 = 31$ units per hour. Team B: $198 \div 6 = 33$. Team C: $240 \div 8 = 30$. Team B completed the most units per hour.

GED-MATH12-012 — Correct: B,D — Correct Answer: $21x - 14 - 4x - 20$ | $7(3x) - 7(2) - 4x - 20$ — Explanation: Both choices correctly show the distributed form before combining like terms.

GED-MATH12-013 — Correct: A — Correct Answer: 21/50 — Explanation: There are 100 participants total. Phone or tablet users total $27 + 15 = 42$, and $42/100$ simplifies to $21/50$.

GED-MATH12-014 — Correct: 9 — Tolerance: exact — Correct Answer: 9 — Explanation: Substitute $a = 18$: $252 = 12q + 8(18)$, so $252 = 12q + 144$. Then $108 = 12q$, so $q = 9$.

GED-MATH12-015 — Correct: C — Correct Answer: 242 square feet — Explanation: Rectangle area is $18 \times 11 = 198$. Triangle area is $1/2(11)(8) = 44$. Total area is $198 + 44 = 242$.

GED-MATH12-016 — Correct: B — Correct Answer: $y = -3x + 19$ — Explanation: When x increases by 6, y decreases by 18, so the rate is -3. Substituting $x = 2$ and $y = 13$ gives $y = -3x + 19$.

GED-MATH12-017 — Correct: A,B — Correct Answer: The medium bag costs $1.60 per pound. | The large bag costs $1.50 per pound. — Explanation: Medium: 28.80 ÷ 18 = $1.60 per pound. Large: 48.00 ÷ 32 = $1.50 per pound.

GED-MATH12-018 — Correct: B — Correct Answer: $3(x + 4) \geq 45$ — Explanation: The sum of a number and 4 is $x + 4$. Three times that sum is $3(x + 4)$, and “at least” means $\geq$.

GED-MATH12-019 — Correct: B — Correct Answer: $160.00 — Explanation: A 15% markup means the selling price is 115% of wholesale cost. So 184 ÷ 1.15 = 160.

GED-MATH12-020 — Correct: 12 — Tolerance: exact — Correct Answer: 12 — Explanation: $3(5x - 4) - 2(4x + 7) = 15x - 12 - 8x - 14 = 7x - 26$. Then $7x - 26 = 58$, so $x = 12$.

GED-MATH12-021 — Correct: B — Correct Answer: 5 feet — Explanation: Area = πr^2. So $78.5 = 3.14r^2$, giving $r^2 = 25$ and $r = 5$.

GED-MATH12-022 — Correct: B — Correct Answer: 3/2 — Explanation: Rate of change = $(12 - (-6))/(4 - (-8)) = 18/12 = 3/2$.

GED-MATH12-023 — Correct: A — Correct Answer: 21.7 — Explanation: The driver has traveled $64.8 + 38.5 = 103.3$ miles. The remaining distance is $125 - 103.3 = 21.7$ miles.

GED-MATH12-024 — Correct: A — Correct Answer: Function A has a greater rate of change than Function B. — Explanation: Function A has rate $(8 - (-1))/(1 - (-2)) = 9/3 = 3$. Function B has rate 2.5, so Function A has the greater rate.

GED-MATH12-025 — Correct: 89.1 — Tolerance: exact — Correct Answer: 89.1 — Explanation: Certification score = $0.30(82) + 0.45(90) + 0.25(96) = 24.6 + 40.5 + 24 = 89.1$.

GED-MATH12-026 — Correct: B — Correct Answer: $b + f = 138$
$18b + 26f = 3{,}084$ — Explanation: The first equation represents the number of tickets, and the second represents total revenue.

GED-MATH12-027 — Correct: C — Correct Answer: 117.6 square inches — Explanation: Original area is $12 \times 5 = 60$. The enlarged area is $60 \times 1.4^2 = 60 \times 1.96 = 117.6$.

GED-MATH12-028 — Correct: D1=−8/5; D2=6 — Correct Answer: D1=−8/5; D2=6 — Explanation: $8x + 5y = 30$ gives $5y = -8x + 30$, so $y = -8/5x + 6$.

GED-MATH12-029 — Correct: B — Correct Answer: 14/95 — Explanation: There are 20 beads total and 8 are clear. Without replacement, P(two clear) = $8/20 \times 7/19 = 14/95$.

GED-MATH12-030 — Correct: A,B — Correct Answer: $s = (C - 2\pi r)/2$ | $s = C/2 - \pi r$ — Explanation: From $C = 2\pi r + 2s$, subtract $2\pi r$ and divide by 2. This is equivalent to $s = C/2 - \pi r$.

GED-MATH12-031 — Correct: C — Correct Answer: 327.6 grams — Explanation: Mass = density × volume = $7.8 \times 42 = 327.6$ grams.

GED-MATH12-032 — Correct: C — Correct Answer: 12 — Explanation: $6(2r - 5) - 3(r + 4) = 12r - 30 - 3r - 12 = 9r - 42$. Then $9r - 42 = 66$, so $r = 12$.

GED-MATH12-033 — Correct: B — Correct Answer: 65.94 in^3 — Explanation: $V = 1/3\pi r^2h = 1/3(3.14)(3^2)(7) = 65.94$ cubic inches.

GED-MATH12-034 — Correct: B — Correct Answer: 4 — Explanation: $x^2 - 13x + 36$ factors as $(x - 4)(x - 9)$, so $x = 4$ is one solution.

GED-MATH12-035 — Correct: A — Correct Answer: The value decreases by 12.5%. — Explanation: Multiplying by 0.875 means 87.5% remains each year, so the decrease is 12.5%.

GED-MATH12-036 — Correct: C — Correct Answer: 63 feet — Explanation: The scale is $27 \div 6 = 4.5$ feet per centimeter. Then $14 \times 4.5 = 63$ feet.

GED-MATH12-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: T1=$y = 3(x - 4) + 8$; T2=$y = 3x - 4 + 8$; T3=$y = 3(x + 8) - 4$ — Explanation: Each tile follows the order of operations described in the target.

GED-MATH12-038 — Correct: C — Correct Answer: 455 — Explanation: The rate is $156 \div 24 = 6.5$ labels per minute. In 70 minutes, the machine prints $6.5 \times 70 = 455$ labels.

GED-MATH12-039 — Correct: A — Correct Answer: $36 + 12g \leq 180$ — Explanation: The total number of people is the 36 staff members plus 12 people for each invited group, and the room can hold no more than 180 people.

GED-MATH12-040 — Correct: A — Correct Answer: The population increases by 18% each time period. — Explanation: Multiplying by 1.18 means the population keeps 100% and increases by 18%.

GED-MATH12-041 — Correct: A,B — Correct Answer: The East route averaged 40 miles per hour. | The West route averaged 45 miles per hour. — Explanation: East: $96 \div 2.4 = 40$ mph. West: $135 \div 3.0 = 45$ mph.

GED-MATH12-042 — Correct: B — Correct Answer: −6 — Explanation: $f(-2) = -2(-2)^2 + 5(-2) + 12 = -8 - 10 + 12 = -6$.

GED-MATH12-043 — Correct: C — Correct Answer: It is not linear because the changes in y are not constant. — Explanation: The changes in y are −5, −3, and −1, so the relationship is not linear.

GED-MATH12-044 — Correct: B — Correct Answer: $1/4(x - 12) + 9$ — Explanation: The difference of x and 12 is $x - 12$. One fourth of that difference is $1/4(x - 12)$, and 9 more is $1/4(x - 12) + 9$.

GED-MATH12-045 — Correct: 28 — Tolerance: exact — Correct Answer: 28 — Explanation: $168 = 1/2(14 + b)(8)$, so $168 = 4(14 + b)$. Then $42 = 14 + b$, so $b = 28$.

GED-MATH12-046 — Correct: C — Correct Answer: 1,836 square feet — Explanation: Parking area is $54 \times 36 = 1{,}944$ square feet. Shed area is $12 \times 9 = 108$ square feet. Uncovered area is $1{,}944 - 108 = 1{,}836$ square feet.